

Supplementary Material for Auto-Decte: Trust-Constrained Document Intelligence

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1 Artifact inventory

The package contains raw per-case predictions, aggregate recognition results, resilience timings, fault-injection outcomes, generated plots and tables, and a manifest with SHA-256 hashes. The manifest records benchmark identifier **eswa-v1**, five deterministic seeds, package versions, source commit, commands, and source-tree cleanliness at generation time.

Artifact	Purpose
<code>artifacts/recognition_raw.json</code>	13,050 model-case rows
<code>artifacts/recognition_summary.json</code>	split, threshold, baseline, robustness metrics
<code>artifacts/trust_faults.json</code>	five injected fault outcomes
<code>artifacts/resilience.json</code>	checks and 30 latency repetitions
<code>artifacts/manifest.json</code>	hashes, environment, commands, source identity

2 Threshold protocol

Calibration uses seeds 101 and 202 (1,740 rows). Evaluation uses seeds 303, 404, and 505 (2,610 rows). The evaluation rows comprise 29 conditions applied to 90 seed-font-digit base units and are therefore not mutually independent. Training seeds for the HOG-SVM baseline are separate. Thresholds from 0 to 0.10 in increments of 0.005 are evaluated; the operating threshold is frozen from the calibration-risk target before evaluation.

3 Perturbation-level outcomes

Table 1 reports the complete generated aggregation at $\tau = 0.08$. Each table row applies one condition to the same 90 evaluation base units. Accepted accuracy is conditional on acceptance and must be interpreted with the reported coverage; the table does not provide 90 new independent units per condition.

4 Fault definitions

Wrong recognition and wrong AI suggestion inject incorrect values through their respective candidate services. Irrelevant retrieval supplies a non-matching reference. Stale human replay submits an outdated expected version. Duplicate evidence submits identical content twice. Passing requires an unchanged pre-existing fact and complete audit evidence; the last two conditions also require their designated exceptions.

5 Reproduction

From the experiment repository and its locked Python environment, run:

Table 1: Coverage and conditional accepted accuracy by perturbation.

Condition	Severity	Coverage (%)	Accepted accuracy (%)
clean	0.00	13.3	100.0
gaussian_blur	0.25	11.1	100.0
gaussian_blur	0.50	14.4	100.0
gaussian_blur	0.75	14.4	100.0
gaussian_blur	1.00	15.6	100.0
gaussian_noise	0.25	12.2	100.0
gaussian_noise	0.50	11.1	100.0
gaussian_noise	0.75	12.2	100.0
gaussian_noise	1.00	12.2	100.0
brightness	0.25	13.3	100.0
brightness	0.50	12.2	100.0
brightness	0.75	12.2	100.0
brightness	1.00	12.2	100.0
rotation	0.25	12.2	100.0
rotation	0.50	13.3	100.0
rotation	0.75	12.2	100.0
rotation	1.00	10.0	100.0
perspective	0.25	8.9	100.0
perspective	0.50	7.8	100.0
perspective	0.75	4.4	100.0
perspective	1.00	3.3	100.0
jpeg	0.25	13.3	100.0
jpeg	0.50	13.3	100.0
jpeg	0.75	11.1	100.0
jpeg	1.00	13.3	100.0
occlusion	0.25	12.2	100.0
occlusion	0.50	12.2	100.0
occlusion	0.75	6.7	100.0
occlusion	1.00	3.3	100.0

```
python -m benchmarks.run_all --config benchmarks/config/eswa-v1.json \
    --output artifacts/eswa-v1
pytest -q
ruff check .
mypy app benchmarks tests/benchmarks
```

Tesseract was unavailable in the recorded environment and is therefore excluded from the quantitative comparison.